

Computer Networks



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Computer Networks

Course Directions:

- network types
- communication protocols (TCP/IP),
- network architecture models
- client/server paradigm,
- BSD socket interface,
- Winsock interface,
- application protocols (terminal, SMTP, FTP, POP, et.al.),
- RPC paradigm,
- peer-to-peer(P2P) paradigm,
- wireless networks,
- security aspects in computer networks

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Bibliography:

- ... (each course)
- Larry L. Peterson , Bruce S. Davie, Computer Networks: A Systems Approach, 6th Edition, 01 Oct 2020
- Lewis Van Winkle, Hands-On Network Programming with C: Learn socket programming in C and write secure and optimized network code, 1st Edition (May 13, 2019)
- Andrew S. Tanenbaum, David J. Wetherall, Computer Networks (5th Edition), ISBN-10: 0132126958 , Publication Date: October 7, 2010
- James F. Kurose, Keith W. Ross; Computer Networking: A Top-Down Approach (6th Edition), 2013 (<http://www-net.cs.umass.edu/kurose-ross-ppt-6e/>)

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Laboratory:

- UNIX/Linux system programming in C
- Communication among processes running on the same computers (signals, pipes, FIFOs, descriptors duplication)
- Communication between processes laying on different computers
 - Iterative / Concurrent servers
 - I/O multiplexing
 - Exception handling communication

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- The main focus is centered on computer network programming (Internet application programming)
- The network hardware part is approached at an informative level

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It requires knowledge about:

- Computer Architecture
- Operating systems
- Programming language: C/C++

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Evaluation

- Final mark(N)
 $N = 0.5 * P + 0.4 * L + 1$

Where:

- P – the project;
 - Types: A (maximum 10), B (maximum 8), C (maximum 6)
- L – laboratory mark;

Calculated as it follows:

- Mandatory problems during the semester;
 - 4th & 10th weeks;
- Individual activities;
- Other laboratory activities;

Conditions: getting a minimum of 5 for P and a minimum of 5 for N

The final mark is computed in accordance with ECTS

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Details:

- Discipline site:
 - <http://profs.info.uaic.ro/~computernetworks>
- Discipline team:
 - Lenuța Alboaie – **course, laboratory**
 - Andrei Panu – **course, laboratory**
 - Georgiana Calancea – **laboratory**
 - Ioana Bogdan – **laboratory**
 - Emanuel Achirei – **laboratory**
 - Madalin Matei - **laboratory**
 - Andrei Scutelnicu – **laboratory**
 - Ecaterina Hrib – **laboratory**
 - Stefan Onu Branisteanu - **laboratory**

Questions?

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FAQ answers:

- It will focus on conceptual understanding and problem solving skills
- ...

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“Everyone is a genius. But if you judge a fish on its ability to climb a tree, it will live its whole life believing it is stupid.” (A. Einstein)

```
server.sin_family = AF_INET;  
server.sin_port = htons (atoi (argv[2]));  
memcpy (&server.sin_addr.s_addr,  
ip_addr->h_addr, sizeof (ip_addr->h_addr));  
client.sin_family = AF_INET;  
client.sin_port = 0;  
client.sin_addr.s_addr = htonl (INADDR_ANY);
```

... ???

The answer: Course &
Laboratory

